

Practice

Lesson 4 Assessment

Loops – Build a Persistent Thinking System (CLI Notes v1)

Topics Covered

- Why loops are needed
 - `while` loops
 - Loop conditions
 - Loop variables and counters
 - Infinite loops
 - User-controlled repetition
 - Menu-driven programs
 - Building a persistent CLI Notes system
 - Basic program state management
-

Part A. Single Choice Questions (10 pts)

1. What is the main purpose of a loop?

- A. Store data permanently
- B. Repeat a block of code
- C. Create variables
- D. Compare two values

Answer:

2. How many times will the loop execute?

```
int count =1;

while (count<=3)
{
    cout<<count<<endl;
    count++;
}
```

- A. 2

- B. 3
- C. 4
- D. Infinite times

Answer:

3. Which part of a `while` loop determines whether the loop continues?

- A. The loop body
- B. The variable declaration
- C. The condition
- D. The output statement

Answer:

4. What is the value of `x` after the loop finishes?

```
int x =0;

while (x<5)
{
    x++;
}
```

- A. 4
- B. 5
- C. 6
- D. Infinite loop

Answer:

5. Which situation is most appropriate for using a loop?

- A. Printing a welcome message once
- B. Checking a condition one time
- C. Displaying a menu until the user chooses Exit
- D. Declaring a variable

Answer:

6. What is an infinite loop?

- A. A loop that runs exactly 100 times
- B. A loop that never executes
- C. A loop that never stops
- D. A loop with multiple conditions

Answer:

7. What will be printed?

```
int n = 3;

while (n>0){
    cout<<"*";
    n--;
}
```

- A. *
- B. **
- C. ***
- D. ****

Answer:

8. Which variable is commonly used to track loop progress?

- A. Counter variable
- B. String variable
- C. Constant variable
- D. Boolean literal

Answer:

9. Why does a CLI Notes application typically use a loop?

- A. To repeatedly display the menu and process commands
- B. To make variables larger
- C. To improve compilation speed
- D. To remove all user input

Answer:

10. Which of the following conditions is most likely to create an infinite loop?

A.

```
while (x<10)
{x++;}
```

B.

```
while (x>0)
```

```
{x--;}
```

C.

```
while (true)
{}
```

D.

```
while (choice==0)
{choice =1;}
```

Answer:

Part B. Fill in the Blanks (10 pts)

1. A loop allows a program to _____ code multiple times.

Answer:

2. The variable that changes during loop execution is often called a _____.

Answer:

3. Complete the code:

```
int i =1;

while(i<=5)
{
    i_____;
}
```

Answer:

4. A loop that never stops is called an _____ loop.

Answer:

Part C. True or False (10 pts)

1. A loop can reduce code duplication.

Answer:

2. A **while** loop always executes at least once.

Answer:

3. If a loop condition is initially false, the loop body will not execute.

Answer:

4. The condition of a `while` loop is checked before each iteration.

Answer:

5. A loop variable should usually change inside the loop.

Answer:

6. The following code creates an infinite loop:

```
int x =1;

while(x>0)
{
}
```

Answer:

7. A menu-driven application often runs inside a loop.

Answer:

8. A loop condition must evaluate to a Boolean value.

Answer:

9. The statement `i++` decreases the value of `i`.

Answer:

10. Loops are useful for building interactive command-line tools.

Answer:

Part D. Code Reading (10 pts) Question 1

What is the output?

```
int i =2;

while(i<=4)
{
    cout<<i<<" ";
}
```

```
i++;  
}
```

Answer

Question 2

What is the final value of `count` ?

```
int count =10;  
  
while(count>7)  
{  
    count--;  
}
```

Answer

Question 3

How many times will the loop execute?

```
int x =0;  
  
while(x<8)  
{  
    x+=2;  
}
```

Answer

Question 4

What is printed?

```
bool running =false;  
  
while(running)  
{  
    cout<<"Hello";  
}
```

Answer

Question 5

What is the output?

```
int notes =1;  
  
while(notes<=3)  
{  
    cout<<"Saving Note"<<endl;
```

```
notes++;  
}
```

Answer

Part E. Short Answer Questions (10 pts)

1. Why are loops important when building interactive software such as CLI Notes?

Sample Answer

2. Explain why the following code causes an infinite loop.

```
inti =1;  
  
while(i<5)  
{  
    cout<<i;  
}
```

Sample Answer

3. What is the role of a loop condition?

Sample Answer

Part F. Programming Questions (20 pts)

Question 1 — Countdown Tool

Write a program that prints:

```
5  
4  
3  
2  
1  
Blast Off!
```

using a `while` loop.

Question 2 — Note Creation Simulator

Ask the user:

```
How many notes would you like to create?
```

Then print:

```
Creating Note 1  
Creating Note 2  
...
```

until all notes have been created.

Example

Input:

```
3
```

Output:

```
Creating Note 1  
Creating Note 2  
Creating Note 3
```

Question 3 — Persistent Menu System

Create a menu that repeatedly displays:

```
1. Add Note  
2. View Notes  
3. Exit
```

The menu should continue appearing until the user chooses `3`.

Question 4 — Password Retry System

Store the correct password:

```
string password = "cpp123";
```

Ask the user to enter a password repeatedly until the correct password is entered.

Output:

```
Access Granted
```

when successful.

Question 5 — CLI Notes v1 (Challenge)

Create a simple note-taking system.

Requirements:

- Use a loop.
- Display a menu:
 - Add Note
 - Show Current Note
 - Exit
- Store one note in a string variable.
- Continue running until Exit is selected.

Example Session

```
1. Add Note
2. Show Current Note
3. Exit
```

```
Choice: 1
Enter note:
Learn C++ loops
```

```
Choice: 2
Current Note:
Learn C++ loops
```

```
Choice: 3
Goodbye!
```

Part G. Thinking Challenge (Optional Bonus)Question

A student writes:

```
int menuChoice =0;

while(menuChoice!=3)
{
    cout<<"1. Add"<<endl;
    cout<<"2. View"<<endl;
    cout<<"3. Exit"<<endl;
}
```

The program never exits.

Why? How can it be fixed?

Sample Answer

Teacher Notes

This assessment focuses on the central idea of Lesson 4:

A loop allows a program to keep thinking and responding instead of running only once.

The progression is intentional:

1. Understanding repetition.
2. Understanding loop conditions.
3. Avoiding infinite loops.
4. Building menu-driven programs.
5. Creating a persistent CLI Notes system.

This naturally prepares students for Lesson 5, where the CLI Notes project can evolve into storing multiple notes using arrays, vectors, or file persistence.